INDEX

NAME: P. Revanth STD: SEC: S6 ROLL NO.: 112191287 SUB: R Programming

S.No.	Date	Title	Page No.	Teacher's Sign / Remarks
		Basic Introduction.		
1.	3/2/26	Learning all the basics of R-Programming.		
2.	10/2/26	operators & Decision making in R-programming.		
3.	17/2/26	learn the basics of functions in R & Implement with example		
4.	17/2/26	Implement data frames in R "write a program joins columns & rows in a data frames:-		
5.	03/3/26	Implement different string Manipulation functions in R.		
6	03/3/26	Implement different data structure in R (vectors, lists, Data frames)		
7	17/3/26	program to read a csv file and analize data in csv file		
8.	24/3/26	creating pie charts & Bar charts using R.		
9.	31/03/26	creating data sets & statistical analysis on data.		

////////////////////////EXP-1//

```
> # Question 1: Student Information System
> roll_no <- 202
> name <- "Rahul"
> cgpa <- 9.1
> pass <- TRUE
> roll_no
[1] 202
> name
[1] "Rahul"
> cgpa
[1] 9.1
> pass
[1] TRUE
>
> # Question 2: Employee Salary Record
> emp_id <- 7002
> salary <- 55000.50
> department <- "HR"
> is_permanent <- FALSE
> emp_id
[1] 7002
> salary
[1] 55000.5
> department
[1] "HR"
> is_permanent
[1] FALSE
>
> # Question 3: Online Shopping Cart
> product_name <- "Mobile"
> price <- 25000
> stock_available <- 12
> discount_eligible <- TRUE
> product_name
[1] "Mobile"
> price
[1] 25000
> stock_available
[1] 12
> discount_eligible
[1] TRUE
```

```
> # Question 4: Weather Monitoring System
> city <- "Hyderabad"
> temperature <- 29.4
> rain <- TRUE
> wind_speed <- 8.6
> city
[1] "Hyderabad"
> temperature
[1] 29.4
> rain
[1] TRUE
> wind_speed
[1] 8.6
>
> # Question 5: Examination Result Processing
> subject <- "Physics"
> total_marks <- 100
> marks_obtained <- 91
> result <- marks_obtained >= 40
> subject
[1] "Physics"
> total_marks
[1] 100
> marks_obtained
[1] 91
> result
[1] TRUE
>
> # Question 6: Banking Application
> account_no <- 987654321
> holder_name <- "Priya"
> balance <- 80000.75
> account_active <- TRUE
> account_no
[1] 987654321
> holder_name
[1] "Priya"
> balance
[1] 80000.75
> account_active
[1] TRUE
```

```
> # Question 7: Movie Rating System
> movie_title <- "Avatar"
> rating <- 9.2
> reviews <- 4500
> available <- TRUE
> movie_title
[1] "Avatar"
> rating
[1] 9.2
> reviews
[1] 4500
> available
[1] TRUE
>
> # Question 8: Traffic Monitoring System
> vehicle_count <- 500
> location <- "City Center"
> avg_speed <- 35.4
> congested <- TRUE
> vehicle_count
[1] 500
> location
[1] "City Center"
> avg_speed
[1] 35.4
> congested
[1] TRUE
```

//////////////////////EXP-2//

```
> # Question 1: Loan Eligibility Check
> age <- 30
> income <- 42000
> if(age > 21 & income > 30000){
+   print("Eligible for Loan")
+ } else {
+   print("Not Eligible for Loan")
+ }
[1] "Eligible for Loan"
>
> # Question 2: Attendance Requirement
> attendance <- 69
> if(attendance >= 75){
+   print("Eligible for Exam")
+ } else {
+   print("Not Eligible for Exam")
+ }
[1] "Not Eligible for Exam"
>
> # Question 3: Online Discount System
> amount <- 6500
> premium_member <- FALSE
> if(amount > 5000 | premium_member){
+   print("Discount Applied")
+ } else {
+   print("No Discount")
+ }
[1] "Discount Applied"
>
> # Question 4: Electricity Bill Slab
> units <- 350
> if(units < 100){
+   print("Low Usage")
+ } else if(units <= 300){
+   print("Medium Usage")
+ } else {
+   print("High Usage")
+ }
[1] "High Usage"
```

```
> # Question 5: Employee Bonus Eligibility
> experience <- 8
> rating <- 4.8
> if(experience > 5 & rating > 4){
+   print("Bonus Eligible")
+ } else {
+   print("Not Eligible for Bonus")
+ }
[1] "Bonus Eligible"
>
> # Question 6: Password Validation
> password <- "mypassword123"
> if(nchar(password) >= 8){
+   print("Valid Password")
+ } else {
+   print("Invalid Password")
+ }
[1] "Valid Password"
>
> # Question 7: Temperature Warning System
> temperature <- -3
> if(temperature < 0 | temperature > 40){
+   print("Extreme Temperature Warning")
+ } else {
+   print("Normal Temperature")
+ }
[1] "Extreme Temperature Warning"
>
> # Question 8: Grading System
> marks <- 95
> if(marks >= 90){
+   grade <- "A"
+ } else if(marks >= 75){
+   grade <- "B"
+ } else if(marks >= 50){
+   grade <- "C"
+ } else {
+   grade <- "Fail"
+ }
> print(grade)
[1] "A"
```

```
> # Question 9: Credit Card Approval System
> age <- 28
> income <- 50000
> credit_score <- 780
> if(age >= 21 & age <= 60 & income >= 25000 & credit_score >= 700){
+   print("Credit Card Approved")
+ } else {
+   print("Credit Card Rejected")
+ }
[1] "Credit Card Approved"
>
> # Question 10: Hospital Patient Risk Classification
> bp <- 130
> sugar <- 180
> if(bp > 140 & sugar > 200){
+   risk <- "High Risk"
+ } else if((bp >= 120 & bp <= 140) | (sugar >= 140 & sugar <= 200)){
+   risk <- "Medium Risk"
+ } else {
+   risk <- "Low Risk"
+ }
> print(risk)
[1] "Medium Risk"
>
> # Question 11: Smart Traffic Signal Control
> vehicle_count <- 250
> if(vehicle_count > 200){
+   print("Long Green Signal")
+ } else if(vehicle_count >= 100){
+   print("Medium Green Signal")
+ } else {
+   print("Short Green Signal")
+ }
[1] "Long Green Signal"
```

```r
# Question 12: Employee Performance Appraisal System
attendance <- 94
performance <- 9
project_completed <- TRUE
if(attendance >= 90 & performance >= 8 & project_completed){
  print("Eligible for Appraisal")
} else {
  print("Not Eligible for Appraisal")
}
] "Eligible for Appraisal"

# Question 13: Online Exam Proctoring System
face_detected <- FALSE
multiple_faces <- FALSE
internet_connected <- TRUE
if(!face_detected | multiple_faces | !internet_connected){
  print("Exam Disqualified")
} else {
  print("Exam Allowed")
}
] "Exam Disqualified"

# Question 14: Dynamic Pricing in Ride-Hailing App
high_demand <- TRUE
available_drivers <- 20
if(high_demand & available_drivers < 50){
  print("Surge Pricing Applied")
} else {
  print("Normal Pricing")
}
] "Surge Pricing Applied"
```

```
> # Question 15: Industrial Machine Safety System
> temperature <- 85
> pressure <- 100
> vibration <- 5
> if(temperature > 80 | pressure > 120 | vibration > 7){
>   print("Machine Shutdown")
> } else {
>   print("Machine Running Normally")
> }
[1] "Machine Shutdown"
>
> # Question 16: University Admission Screening
> entrance_score <- 65
> twelfth_marks <- 80
> age <- 21
> if(entrance_score >= 70 & twelfth_marks >= 75 & age <= 25){
>   print("Admission Granted")
> } else {
>   print("Admission Rejected")
> }
[1] "Admission Rejected"
```

//////////////////////////EXP-3//

```
> # Question 1: Bank Interest Calculator
> simple_interest <- function(p, r, t){
-   si <- (p * r * t) / 100
-   return(si)
- }
> simple_interest(15000, 6, 3)
[1] 2700
>
> # Question 2: Student Result Processing
> check_result <- function(marks){
-   if(marks >= 40){
-     return("Pass")
-   } else {
-     return("Fail")
-   }
- }
> check_result(39)
[1] "Fail"
>
> # Question 3: Employee Salary Calculation
> net_salary <- function(basic_salary){
-   tax <- basic_salary * 0.10
-   net <- basic_salary - tax
-   return(net)
- }
> net_salary(45000)
[1] 40500
>
> # Question 4: Electricity Bill Generator
> electricity_bill <- function(units){
-   if(units <= 100){
-     bill <- units * 2
-   } else if(units <= 200){
-     bill <- (100 * 2) + (units - 100) * 3
-   } else {
-     bill <- (100 * 2) + (100 * 3) + (units - 200) * 5
-   }
-   return(bill)
- }
> electricity_bill(180)
[1] 440
```

```
> # Question 5: Online Shopping Discount
> final_amount <- function(amount){
+   if(amount >= 5000){
+     discount <- amount * 0.20
+   } else if(amount >= 2000){
+     discount <- amount * 0.10
+   } else {
+     discount <- 0
+   }
+   return(amount - discount)
+ }
> final_amount(5200)
[1] 4160
>
> # Question 6: Temperature Conversion System
> celsius_to_fahrenheit <- function(c){
+   f <- (c * 9/5) + 32
+   return(f)
+ }
> celsius_to_fahrenheit(25)
[1] 77
>
> # Question 7: Grading System
> assign_grade <- function(marks){
+   if(marks >= 90){
+     "A"
+   } else if(marks >= 75){
+     "B"
+   } else if(marks >= 50){
+     "C"
+   } else {
+     "Fail"
+   }
+ }
> assign_grade(49)
[1] "Fail"
```

```
> # Question 8: Voting Eligibility Checker
> check_voting <- function(age){
+   if(age >= 18){
+     return("Eligible to Vote")
+   } else {
+     return("Not Eligible to Vote")
+   }
+ }
> check_voting(21)
[1] "Eligible to Vote"
>
> # Question 9: Bank Loan EMI Calculator System
> calculate_emi <- function(principal, rate, years){
+   monthly_rate <- rate / (12 * 100)
+   months <- years * 12
+   emi <- (principal * monthly_rate * (1 + monthly_rate)^months) /
+     ((1 + monthly_rate)^months - 1)
+   return(round(emi, 2))
+ }
> calculate_emi(800000, 8.5, 15)
[1] 7877.92
>
> # Question 10: Hospital Patient Billing System
> hospital_bill <- function(consultation, room_charge, days){
+   subtotal <- consultation + (room_charge * days)
+   tax <- subtotal * 0.05
+   total <- subtotal + tax
+   return(total)
+ }
> hospital_bill(1000, 2500, 4)
[1] 11550
```

```
> # Question 11: Online Shopping Order Processing System
> order_total <- function(cart_value){
+   if(cart_value >= 3000){
+     discount <- cart_value * 0.10
+   } else {
+     discount <- 0
+   }
+   if(cart_value >= 5000){
+     delivery <- 0
+   } else {
+     delivery <- 100
+   }
+   final_amount <- cart_value - discount + delivery
+   return(final_amount)
+ }
> order_total(5500)
[1] 4950
>
> # Question 12: University GPA Calculation System
> calculate_gpa <- function(marks){
+   avg <- mean(marks)
+   if(avg >= 85){
+     result <- "Distinction"
+   } else if(avg >= 70){
+     result <- "First Class"
+   } else if(avg >= 50){
+     result <- "Second Class"
+   } else {
+     result <- "Fail"
+   }
+   return(result)
+ }
> calculate_gpa(c(90, 85, 88, 92, 87))
[1] "Distinction"
>
```

```r
# Question 13: Smart Electricity Meter System
smart_meter_bill <- function(peak_units, offpeak_units){
  bill <- (peak_units * 6) + (offpeak_units * 3)
  total_units <- peak_units + offpeak_units
  if(total_units > 300){
    bill <- bill * 1.10
  }
  return(bill)
}
smart_meter_bill(200, 150)
```

```
1] 1815
```

```r
# Question 14: Insurance Premium Calculation System
insurance_premium <- function(base, age, smoker){
  premium <- base
  if(age > 50){
    premium <- premium * 1.20
  }
  if(smoker){
    premium <- premium * 1.30
  }
  return(premium)
}
insurance_premium(6000, 55, TRUE)
```

```
1] 9360
```

```r
# Question 15: Employee Payroll Processing System
payroll <- function(basic){
  hra <- basic * 0.20
  da <- basic * 0.10
  gross <- basic + hra + da
  tax <- gross * 0.12
  net <- gross - tax
  return(net)
}
payroll(50000)
```

```
] 57200
```

```r
# Question 16: Smart City Water Usage Monitoring
water_usage_status <- function(litres){
  if(litres <= 500){
    "Normal"
  } else if(litres <= 1000){
    "High Usage"
  } else {
    "Critical"
  }
}
water_usage_status(1200)
```

```
] "Critical"
```

////////////////////////////////EXP-4//

```
# Question 1: College Student Record System
students <- data.frame(
  RollNo = c(201, 202, 203),
  Name = c("Rahul", "Priya", "Kiran"),
  Dept = c("CSE", "ECE", "IT")
)

marks <- data.frame(
  Marks = c(88, 91, 79)
)

student_data <- cbind(students, marks)
student_data
RollNo  Name Dept Marks
   201 Rahul  CSE    88
   202 Priya  ECE    91
   203 Kiran   IT    79

new_students <- data.frame(
  RollNo = c(204),
  Name = c("Anjali"),
  Dept = c("CSE"),
  Marks = c(85)
)

final_data <- rbind(student_data, new_students)
final_data
RollNo   Name Dept Marks
   201  Rahul  CSE    88
   202  Priya  ECE    91
   203  Kiran   IT    79
   204 Anjali  CSE    85
```

```r
# Question 2: Employee Management System
employee_details <- data.frame(
  EmpID = c(11, 12, 13),
  Name = c("Aman", "Sneha", "Ravi"),
  Dept = c("HR", "IT", "Finance")
)

salary_details <- data.frame(
  Salary = c(40000, 55000, 60000)
)

employees <- cbind(employee_details, salary_details)
employees
EmpID  Name    Dept Salary
   11  Aman      HR  40000
   12 Sneha      IT  55000
   13  Ravi Finance  60000

new_employee <- data.frame(
  EmpID = 14,
  Name = "Neha",
  Dept = "IT",
  Salary = 58000
)

employees <- rbind(employees, new_employee)
employees
EmpID  Name    Dept Salary
   11  Aman      HR  40000
   12 Sneha      IT  55000
   13  Ravi Finance  60000
   14  Neha      IT  58000
```

```r
# Question 3: Hospital Patient Records
patient_info <- data.frame(
  PatientID = c(301, 302),
  Name = c("Suresh", "Lakshmi"),
  Age = c(50, 40)
)

treatment_cost <- data.frame(
  Cost = c(18000, 22000)
)

patient_records <- cbind(patient_info, treatment_cost)
patient_records
PatientID    Name Age  Cost
      301  Suresh  50 18000
      302 Lakshmi  40 22000

new_patient <- data.frame(
  PatientID = 303,
  Name = "Arjun",
  Age = 45,
  Cost = 20000
)

patient_records <- rbind(patient_records, new_patient)
patient_records
PatientID    Name Age  Cost
      301  Suresh  50 18000
      302 Lakshmi  40 22000
      303   Arjun  45 20000
```

```r
# Question 4: Product Inventory System
products <- data.frame(
  ProductID = c(1, 2, 3),
  ProductName = c("Pen", "Book", "Bottle")
)

stock <- data.frame(
  Quantity = c(100, 60, 40)
)

inventory <- cbind(products, stock)
inventory
ProductID ProductName Quantity
        1         Pen      100
        2        Book       60
        3      Bottle       40

new_product <- data.frame(
  ProductID = 4,
  ProductName = "Bag",
  Quantity = 30
)

inventory <- rbind(inventory, new_product)
inventory
ProductID ProductName Quantity
        1         Pen      100
        2        Book       60
        3      Bottle       40
        4         Bag       30
```

//////////////////////////EXP-5//

```
# Question 1: Employee Name Standardization
names <- c("arun", "meena", "vijay")
upper_names <- toupper(names)
name_length <- nchar(upper_names)
upper_names
1] "ARUN"  "MEENA" "VIJAY"
name_length
1] 4 5 5

# Question 2: Student Email Processing
emails <- c("arun@gmail.com", "meena@yahoo.com", "vijay@outlook.com")
usernames <- sub("@.*", "", emails)
usernames
1] "arun"  "meena" "vijay"

# Question 3: Online Form Input Cleanup
input <- c(" Python ", " Machine Learning ", " Cloud ")
clean_input <- trimws(input)
clean_input
1] "Python"           "Machine Learning" "Cloud"

# Question 4: Product Code Validation
product_codes <- c("PRD555", "PRD777", "XYZ999")
valid_codes <- startsWith(product_codes, "PRD")
valid_codes
1]  TRUE  TRUE FALSE

# Question 5: Text Replacement in Reports
report <- "error in login, error in payment"
updated_report <- gsub("error", "issue", report)
updated_report
1] "issue in login, issue in payment"

# Question 6: Password Strength Checker
password <- "welcome123"
if(nchar(password) >= 8){
  print("Strong Password")
} else {
  print("Weak Password")
}
1] "Strong Password"
```

```
# Question 7: Sentence Word Count
sentence <- "Python is easy to learn"
word_count <- length(strsplit(sentence, " ")[[1]])
word_count
1] 5

# Question 8: File Extension Extraction
files <- c("notes.txt", "video.mp4", "music.mp3")
extensions <- sub(".*\\.", "", files)
extensions
1] "txt" "mp4" "mp3"

# Question 9: Student Name Abbreviation
names <- c("Suresh", "Lakshmi", "Deepika")
abbr <- substr(names, 1, 3)
abbr
1] "Sur" "Lak" "Dee"

# Question 10: Search Operation in Text
names <- c("Arun", "Bala", "Ravi", "John")
matched_names <- grep("a", names, value = TRUE)
matched_names
1] "Bala" "Ravi"
```

////////////////////////////EXP-6//

```
# Question 1: Student Marks Processing (Vectors)
marks <- c(65, 72, 88, 91, 79)
total <- sum(marks)
average <- mean(marks)
highest <- max(marks)
total
L] 395
average
L] 79
highest
L] 91

# Question 2: Temperature Monitoring System (Vectors)
temperature <- c(28, 30, 29, 31, 32, 33, 30)
min_temp <- min(temperature)
max_temp <- max(temperature)
min_temp
L] 28
max_temp
L] 33


# Question 3: Employee Profile Storage (Lists)
employee <- list(
  ID = 205,
  Name = "Arjun",
  Salary = 55000,
  Department = "Finance"
)
employee
ID
L] 205

Name
L] "Arjun"

Salary
L] 55000

Department
L] "Finance"

employee$Name
L] "Arjun"
employee$Salary
L] 55000
```

```
# Question 4: Patient Medical Record System (Lists)
patient <- list(
  PatientID = 305,
  Name = "Meena",
  Age = 50,
  TestResults = c(110, 135, 128)
)
patient
atientID
] 305

ame
] "Meena"

ge
] 50

estResults
] 110 135 128

patient$TestResults
] 110 135 128

# Question 5: College Student Database (Data Frames)
students <- data.frame(
  RollNo = c(11, 12, 13),
  Name = c("Rahul", "Sneha", "Kiran"),
  Dept = c("CSE", "ECE", "IT"),
  Marks = c(78, 85, 92)
)
students
RollNo  Name Dept Marks
    11 Rahul  CSE    78
    12 Sneha  ECE    85
    13 Kiran   IT    92

> # Question 6: Sales Report System (Data Frames)
> sales <- data.frame(
+   Month = c("Apr", "May", "Jun"),
+   Sales = c(45000, 52000, 61000)
+ )
> total_sales <- sum(sales$Sales)
> total_sales
[1] 158000
>
> # Question 7: E-Commerce Order System
> prices <- c(999, 1499, 299)
> customer <- list(
+   CustomerID = 401,
+   Name = "Priya",
+   City = "Hyderabad"
+ )
> orders <- data.frame(
+   Item = c("Mouse", "Keyboard", "USB"),
+   Price = prices
+ )
> prices
[1]  999 1499  299
> customer
$CustomerID
[1] 401

$Name
[1] "Priya"

$City
[1] "Hyderabad"

> orders
      Item Price
1    Mouse   999
2 Keyboard  1499
3      USB   299
>
```

```
# Question 8: Online Exam System
subject_marks <- c(82, 76, 89)
student_profile <- list(
  Name = "Vijay",
  RollNo = 121,
  Marks = subject_marks
)
class_results <- data.frame(
  RollNo = c(111, 112, 113),
  Total = c(250, 240, 270)
)
student_profile
ame
] "Vijay"

ollNo
] 121

arks
] 82 76 89

class_results
RollNo Total
   111   250
   112   240
   113   270

# Question 9: Bank Account Management
balances <- c(8000, 8500, 8200)
customer <- list(
  AccountNo = 56789,
  Name = "Lakshmi"
)
transactions <- data.frame(
  Date = c("10-01", "11-01", "12-01"),
  Amount = c(-1000, 2000, -500)
)
balances
1] 8000 8500 8200
customer
AccountNo
1] 56789

Name
1] "Lakshmi"

transactions
 Date Amount
10-01  -1000
11-01   2000
12-01   -500
```

```
> # Question 10: Real-World Summary Question
> scores <- c(78, 84, 91)
> profile <- list(
+   Name = "Suresh",
+   Age = 22,
+   Scores = scores
+ )
> records <- data.frame(
+   Subject = c("Math", "Science", "English"),
+   Marks = scores
+ )
> profile
$Name
[1] "Suresh"

$Age
[1] 22

$Scores
[1] 78 84 91

> records
  Subject Marks
1    Math    78
2 Science    84
3 English    91
```

//////////////////////////EXP-7//////////////////////////

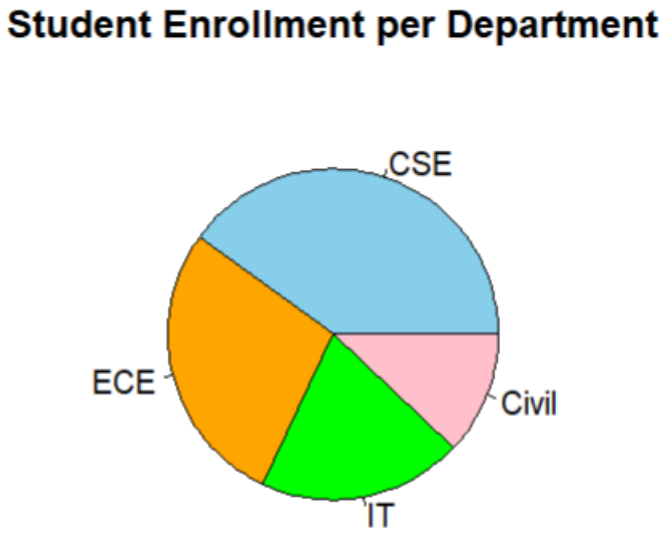
```
# View first few rows
head(sales)
      Date Product    Category Quantity Price
01-01-2026   Phone Electronics        5 20000
01-01-2026     Bag Accessories       10  1500
02-01-2026  Laptop Electronics        3 50000
02-01-2026 Headset Electronics        7  2500
03-01-2026     Bag Accessories        5  1500
03-01-2026   Phone Electronics        3 20000

# Check structure and summary
str(sales)
data.frame':   6 obs. of  5 variables:
$ Date    : chr  "01-01-2026" "01-01-2026" "02-01-2026" "02-01-2026" ...
$ Product : chr  "Phone" "Bag" "Laptop" "Headset" ...
$ Category: chr  "Electronics" "Accessories" "Electronics" "Electronics" ...
$ Quantity: int  5 10 3 7 5 3
$ Price   : int  20000 1500 50000 2500 1500 20000
summary(sales)
    Date             Product            Category           Quantity        Price
Length:6           Length:6           Length:6           Min.   : 3.0   Min.   : 1500
Class :character   Class :character   Class :character   1st Qu.: 3.5   1st Qu.: 1750
Mode  :character   Mode  :character   Mode  :character   Median : 5.0   Median :11250
                                                          Mean   : 5.5   Mean   :15917
                                                          3rd Qu.: 6.5   3rd Qu.:20000
                                                          Max.   :10.0   Max.   :50000
```

```
> # Total Sales per Product
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, sum)
> total_sales_product
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
>
> # Product with Highest Sales
> top_product <- total_sales_product[which.max(total_sales_product$TotalSales), ]
> top_product
  Product TotalSales
4   Phone     160000
>
> # Total Quantity Sold per Category
> quantity_per_category <- aggregate(Quantity ~ Category, data=sales, sum)
> quantity_per_category
     Category Quantity
1 Accessories       15
2 Electronics       18
>
> # Filter High-Value Products
> high_sales <- total_sales_product[total_sales_product$TotalSales > 10000, ]
> high_sales
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
```

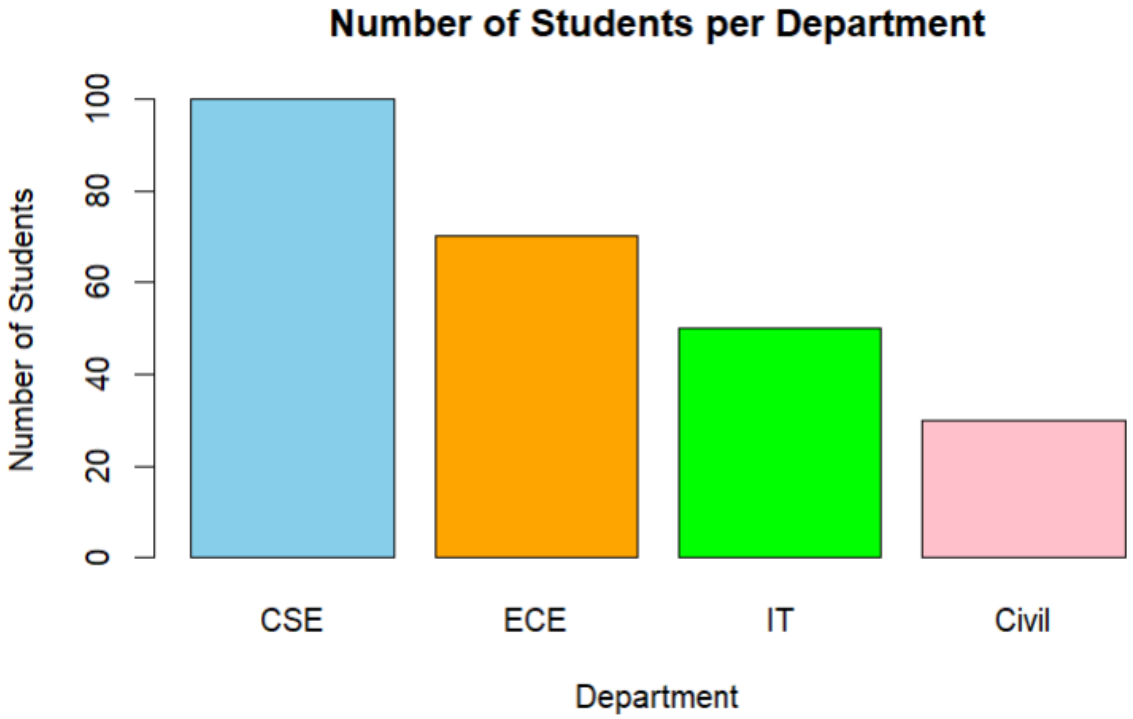
//////////////////////////EXP-8//

 # Question 1: University Course Enrollment

#PIECHART

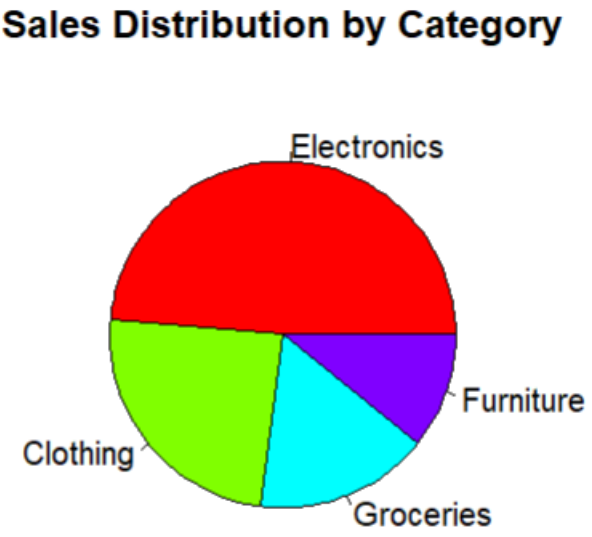


#BARCHART

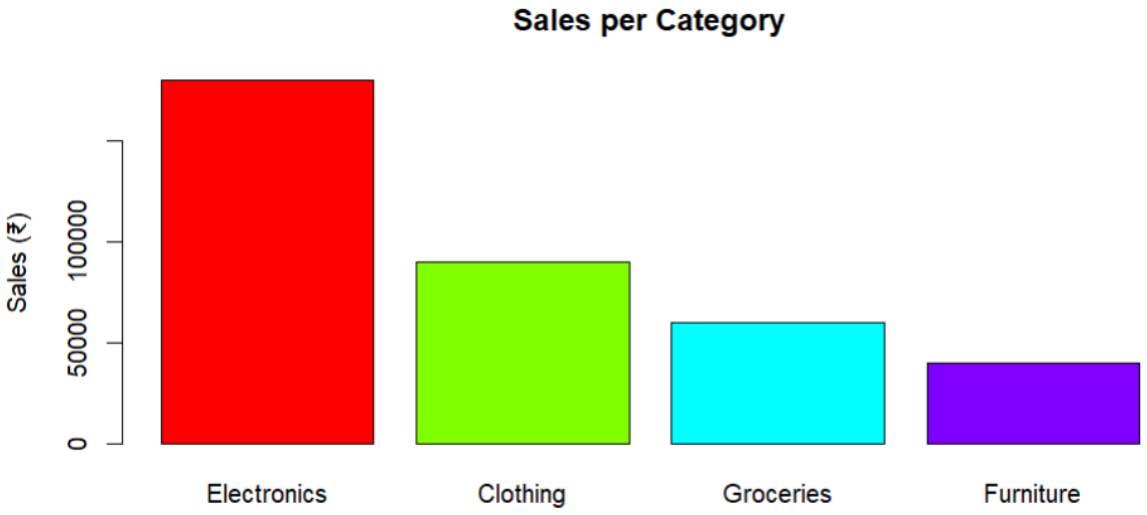


Question 2: Retail Store Sales Distribution

#PIECHART

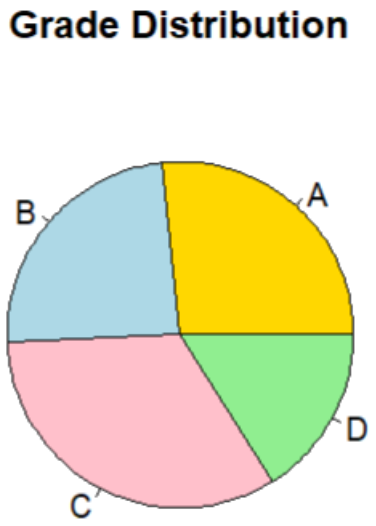


#BARCHART

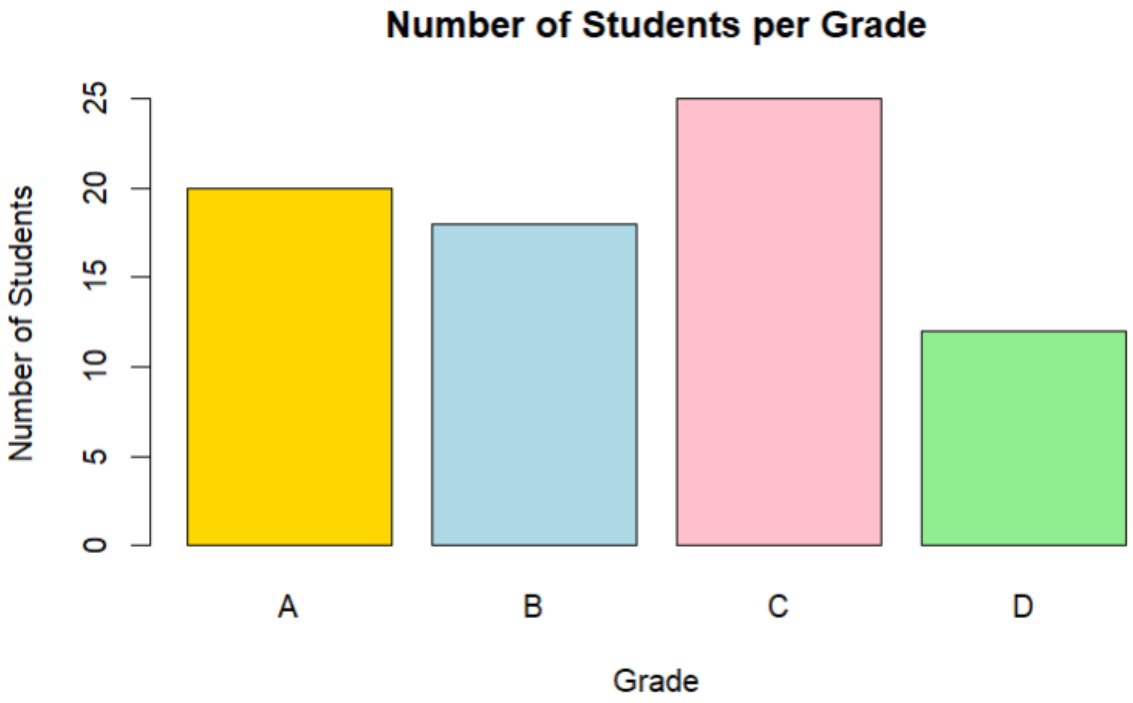


Question 3: Online Exam Result Analysis

#PIECHART

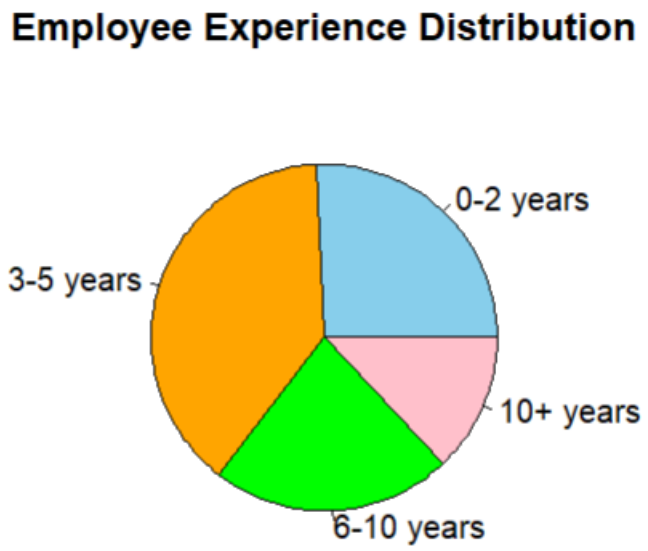


#BARCHART



Question 4: Company Employee Experience Levels

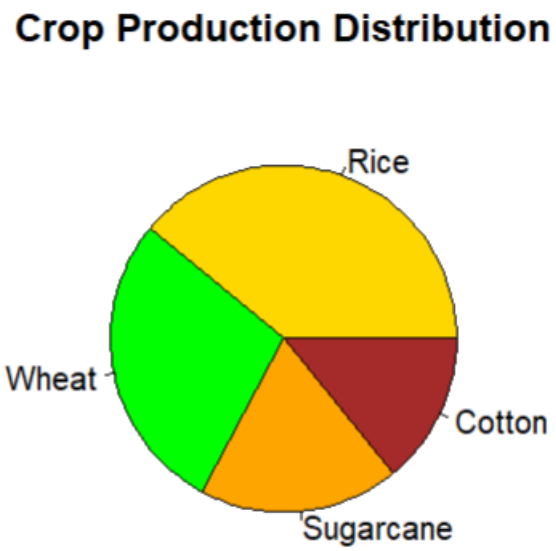
#PIECHART



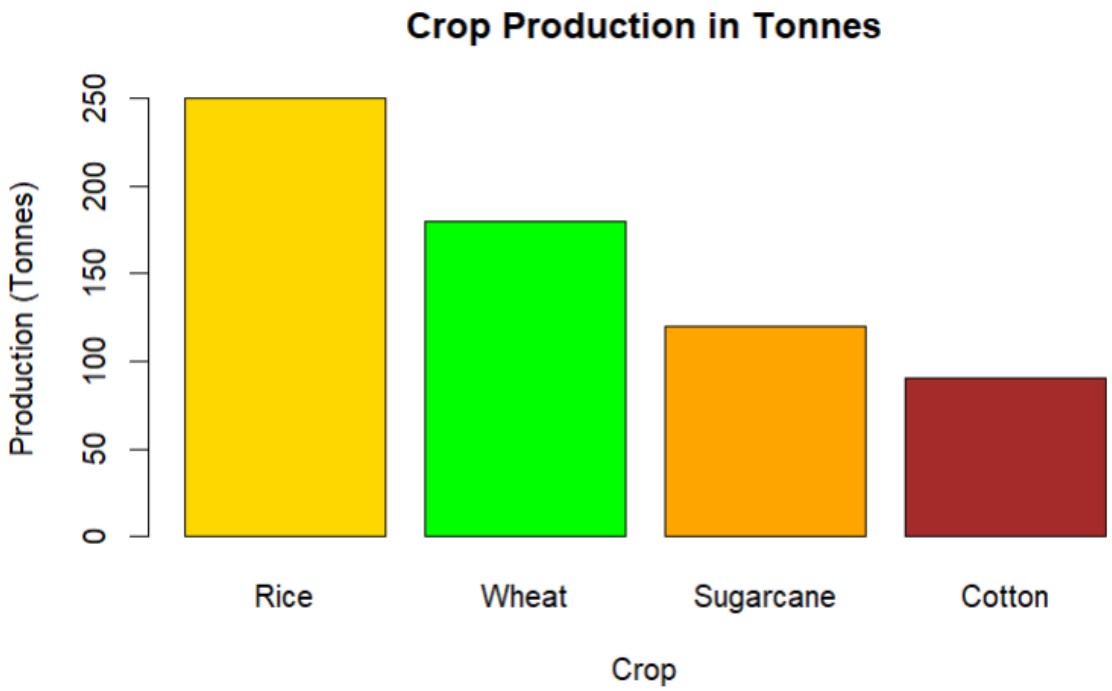
#BARCHART

Question 5: Agricultural Crop Production

#PIECHART



#BARCHART



//////////////////////EXP-9//

```
> # Question 1: School Student Marks Analysis
>
> students <- data.frame(
+   Name = c("Arun","Priya","Kiran","Meena","Rahul","Sneha","Vijay","Lakshmi","Suresh","Anu"),
+   Maths = c(78,85,90,88,76,92,81,79,95,84),
+   Science = c(80,89,85,90,72,94,78,83,91,86),
+   English = c(82,87,88,91,75,93,80,85,94,89)
+ )
>
> mean_marks <- sapply(students[,2:4], mean)
> median_marks <- sapply(students[,2:4], median)
> sd_marks <- sapply(students[,2:4], sd)
>
> students$Total <- rowSums(students[,2:4])
> top_student <- students[which.max(students$Total), ]
>
> avg_math <- mean(students$Maths)
> above_avg_math <- students[students$Maths > avg_math, ]
>
> mean_marks
  Maths Science English
   84.8    84.8    86.4
> median_marks
  Maths Science English
   84.5    85.5    87.5
> sd_marks
   Maths  Science  English
6.373556 6.713171 6.003703
> top_student
    Name Maths Science English Total
9 Suresh    95      91      94   280
> above_avg_math
    Name Maths Science English Total
2  Priya    85      89      87   261
3  Kiran    90      85      88   263
4  Meena    88      90      91   269
6  Sneha    92      94      93   279
9 Suresh    95      91      94   280
>
```

```
> # Question 2: Retail Sales Data Analysis
>
> sales <- data.frame(
-   Product = c("Mobile","Laptop","Mouse","Bag","Tablet","Camera","Speaker"),
-   Day1 = c(4,3,8,10,5,2,6),
-   Day2 = c(5,2,7,9,6,3,5),
-   Day3 = c(6,4,6,8,7,2,7),
-   Day4 = c(5,3,7,11,6,4,6),
-   Day5 = c(7,5,5,10,8,3,8)
- )
>
> sales_stats <- data.frame(
-   Product = sales$Product,
-   Mean = apply(sales[,2:6], 1, mean),
-   Max = apply(sales[,2:6], 1, max),
-   Min = apply(sales[,2:6], 1, min),
-   Variance = apply(sales[,2:6], 1, var)
- )
>
> max_avg_product <- sales_stats[which.max(sales_stats$Mean), ]
>
> sales_stats
  Product Mean Max Min Variance
  Mobile  5.4   7   4      1.3
  Laptop  3.4   5   2      1.3
   Mouse  6.6   8   5      1.3
     Bag  9.6  11   8      1.3
  Tablet  6.4   8   5      1.3
  Camera  2.8   4   2      0.7
 Speaker  6.4   8   5      1.3
> max_avg_product
  Product Mean Max Min Variance
     Bag  9.6  11   8      1.3
>
```

```
>
> # Question 3: Hospital Patient Age Analysis
>
> patients <- data.frame(
+   PatientID = 301:310,
+   Name = c("Ravi","Meena","Arjun","Sneha","Kiran","Priya","Vijay","Lakshmi","Suresh","Anu"),
+   Age = c(40,55,60,35,48,52,45,38,65,50)
+ )
>
> mean_age <- mean(patients$Age)
> median_age <- median(patients$Age)
> range_age <- range(patients$Age)
> sd_age <- sd(patients$Age)
>
> older_patients <- patients[patients$Age > 50, ]
>
> mean_age
[1] 48.8
> median_age
[1] 49
> range_age
[1] 35 65
> sd_age
[1] 9.647107
> older_patients
  PatientID   Name Age
2       302  Meena  55
3       303  Arjun  60
6       306  Priya  52
9       309 Suresh  65
>
```

```
> # Question 4: Employee Salary Analysis
>
> employees <- data.frame(
+   EmpID = 201:210,
+   Name = c("Arun","Priya","Kiran","Meena","Rahul","Sneha","Vijay","Lakshmi","Suresh","Anu"),
+   Dept = c("IT","HR","Finance","IT","HR","Finance","IT","HR","Finance","IT"),
+   Salary = c(35000,38000,45000,36000,39000,47000,34000,40000,50000,37000)
+ )
>
> dept_stats <- aggregate(Salary ~ Dept, data=employees,
+                         function(x) c(Mean=mean(x), Median=median(x)))
> dept_stats <- do.call(data.frame, dept_stats)
>
> overall_avg <- mean(employees$Salary)
> high_earners <- employees[employees$Salary > overall_avg, ]
>
> dept_stats
     Dept Salary.Mean Salary.Median
1 Finance    47333.33         47000
2      HR    39000.00         39000
3      IT    35500.00         35500
> high_earners
  EmpID   Name    Dept Salary
3   203  Kiran Finance  45000
6   206  Sneha Finance  47000
9   209 Suresh Finance  50000
>
```

```
> # Question 5: Online Store Customer Ratings
>
> ratings <- data.frame(
+   ProductID = 1:10,
+   Product = c("Phone","Laptop","Mouse","Bag","Camera","Speaker","Tablet","Watch","Charger","Keyboard"),
+   Rating = c(5,4,3,5,4,5,4,3,5,4)
+ )
>
> mean_rating <- mean(ratings$Rating)
> median_rating <- median(ratings$Rating)
> var_rating <- var(ratings$Rating)
>
> getmode <- function(v){
+   uniqv <- unique(v)
+   uniqv[which.max(tabulate(match(v, uniqv)))]
+ }
>
> mode_rating <- getmode(ratings$Rating)
>
> top_rated <- ratings[ratings$Rating > 4, ]
>
> mean_rating
[1] 4.2
> median_rating
[1] 4
> mode_rating
[1] 5
> var_rating
[1] 0.6222222
> top_rated
  ProductID Product Rating
1         1   Phone      5
4         4     Bag      5
6         6 Speaker      5
9         9 Charger      5
>
```